

01

COACHING FRAMEWORK

Value of Independence

If the coach were to walk away right now, how well would the learner perform?
 We study how an AI agent can act as a coach to accelerate human motor-skill development.

The Human-AI Coaching Game

Human objective
 Task performance

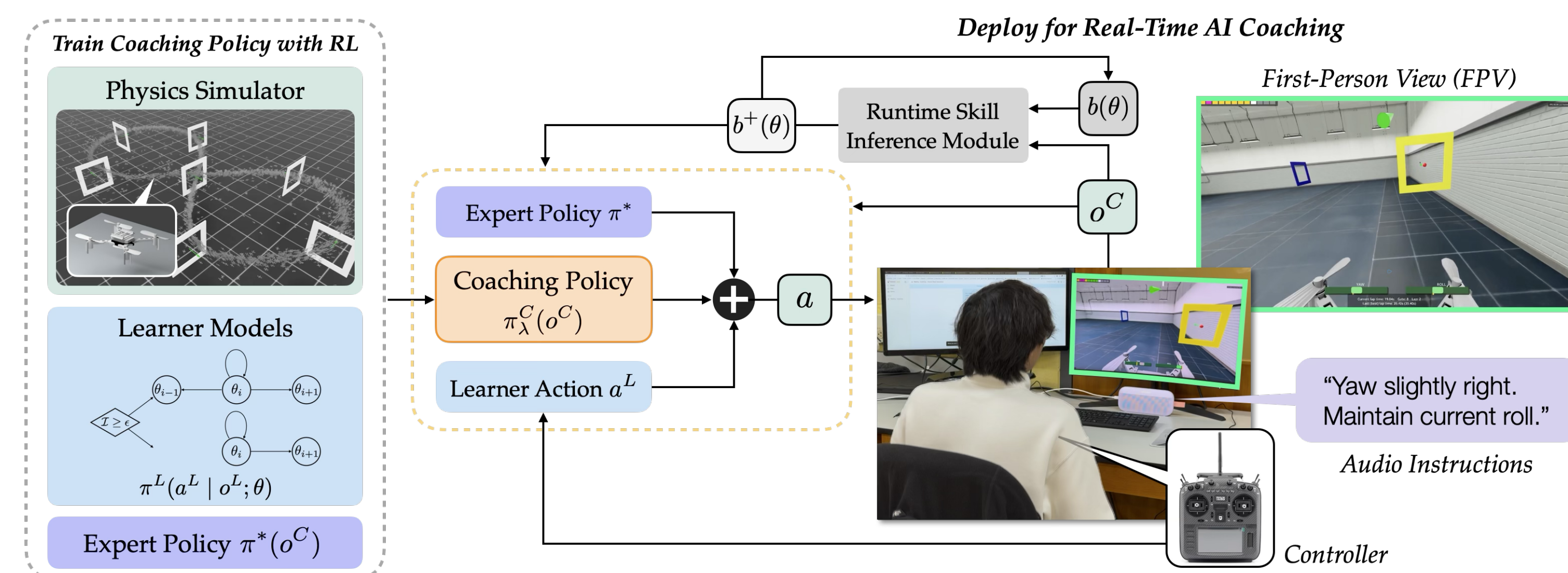
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Coach objective
 Independent competence

We cast the AI coaching problem as a two-player partially observable stochastic game. The learner seeks to maximize task performance, while the coach optimizes a pedagogical objective, namely the learner's *independent competence*.

Learning to Coach (L2C)

The AI coach implements shared control by fusing the learner's action with an expert baseline via a linear blending rule:



$$a_t = \lambda_t \odot \pi^*(o_t^C) + (\mathbf{1} - \lambda_t) \odot a_t^H$$

Our current system learns only the assistance modulation of physical control, while verbal and visual cues are predefined and triggered by fixed rules.

User Study

We conducted a user study with $N = 33$ participants from diverse drone-flying backgrounds in a high-fidelity FPV drone-racing simulator.

Pre-test

2 unassisted laps

Training

15 laps / ~40 min

Post-test

2 unassisted laps

Participants were randomly assigned to L2C, rule-based fading (RBF), or minimally invasive assistance (MIA).

Every third coached lap is unassisted, providing a clean signal for updating the per-gate belief.

All three methods use the same simulator, track, expert policy, language instructions, and visual cues.

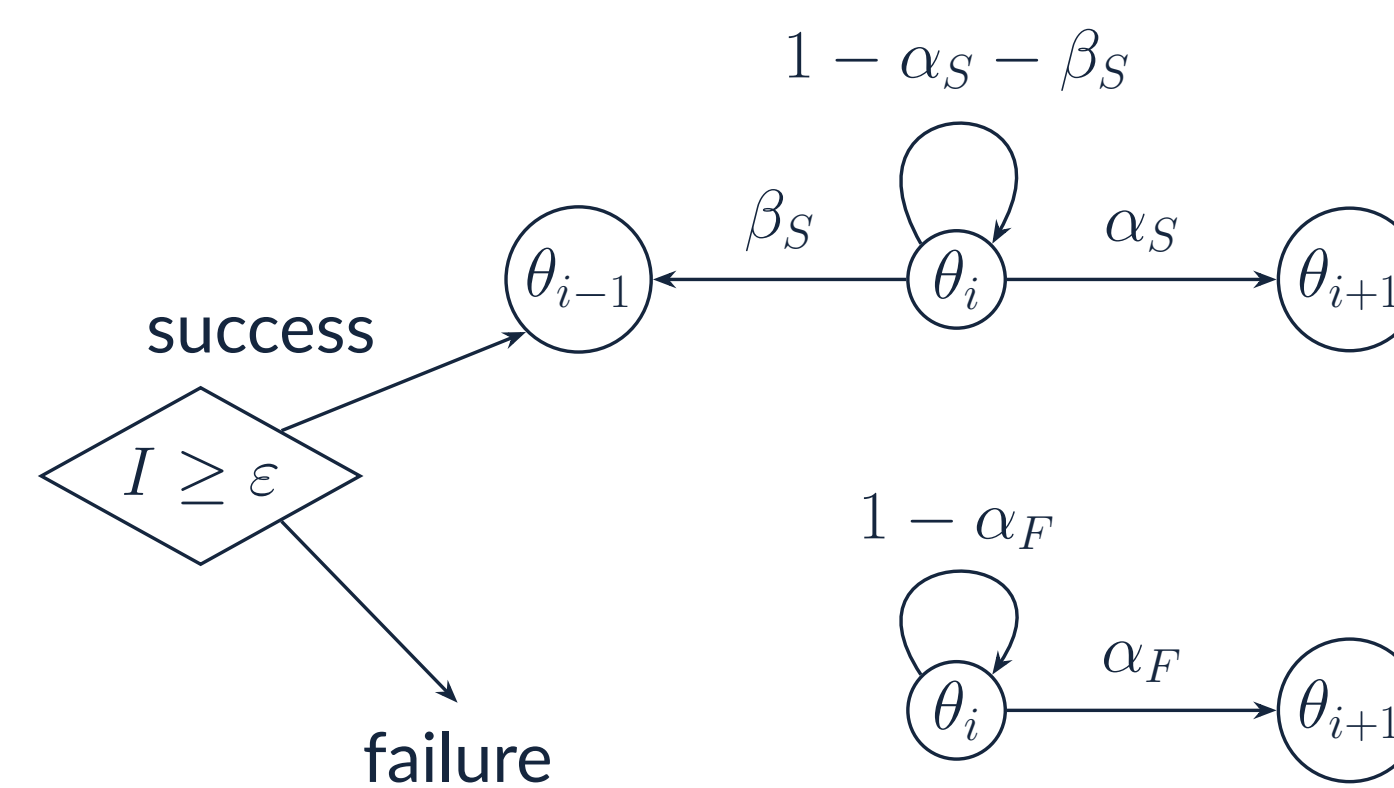
02

SKILL & OBSERVATION

Modeling Skill Progression

$$\pi^H(a^H | o^H; \theta) \propto \exp(\beta(\theta)Q^*(o^H, a^H))$$

A simulated novice samples actions broadly, while more skilled learners concentrate their actions more sharply around the expert action.



When the learner succeeds, they upskill, downskill, or remain unchanged with skill-dependent probabilities; when the learner fails, they may still upskill.

The coach receives the tractable surrogate reward for skill improvement

$$r_t^C = \theta_t - \theta_{t-1}$$

Task Behavior as a Cue for Skill

In drone racing, we partition the track into per-gate segments and maintain an independent belief over the learner's skill at each gate. The per-gate likelihood is a Gaussian on realized segment passage time, centered at a skill-dependent target time calibrated from a pilot study.

$$b^+(\theta) \propto P(o^C | \theta)b(\theta)$$

Segment time
 Observation

Skill belief
 Capability

Assistance
 Control

A fast passage with λ near one is therefore not direct evidence that the learner can repeat the passage alone.

Our study addresses this problem with unassisted evaluation laps. The pre- and post-coaching tests are also unassisted.

A Cue Evaluation Protocol

PROPOSED EVALUATION

1. A cue model for coaching should first be tested on future unassisted behavior.
2. The probe schedule should be fixed across methods. The same initial-skill distribution, task segments, and assistance-policy architecture should be used when cue sets are compared.
3. Calibration should be stratified by gate and assistance level.

Control trajectories, gaze, or verbal feedback can also update the belief through learned likelihoods.

03

EVIDENCE & IMPLICATIONS

Experimental Results

27.9%

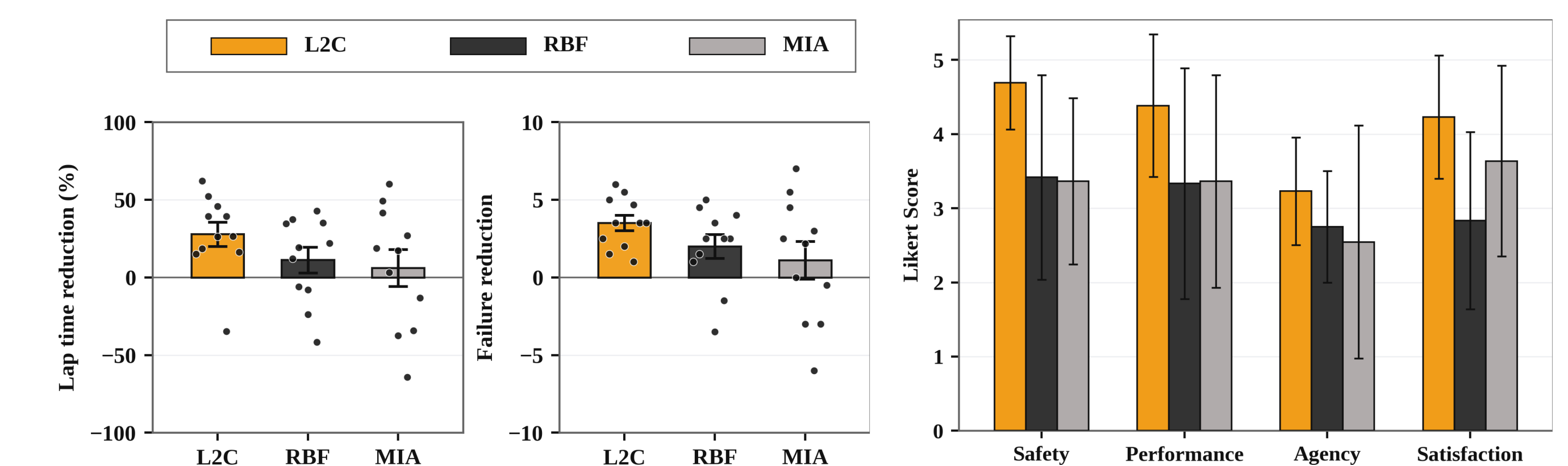
mean lap-time reduction

$p = 0.005, d_z = -1.08$

3.52

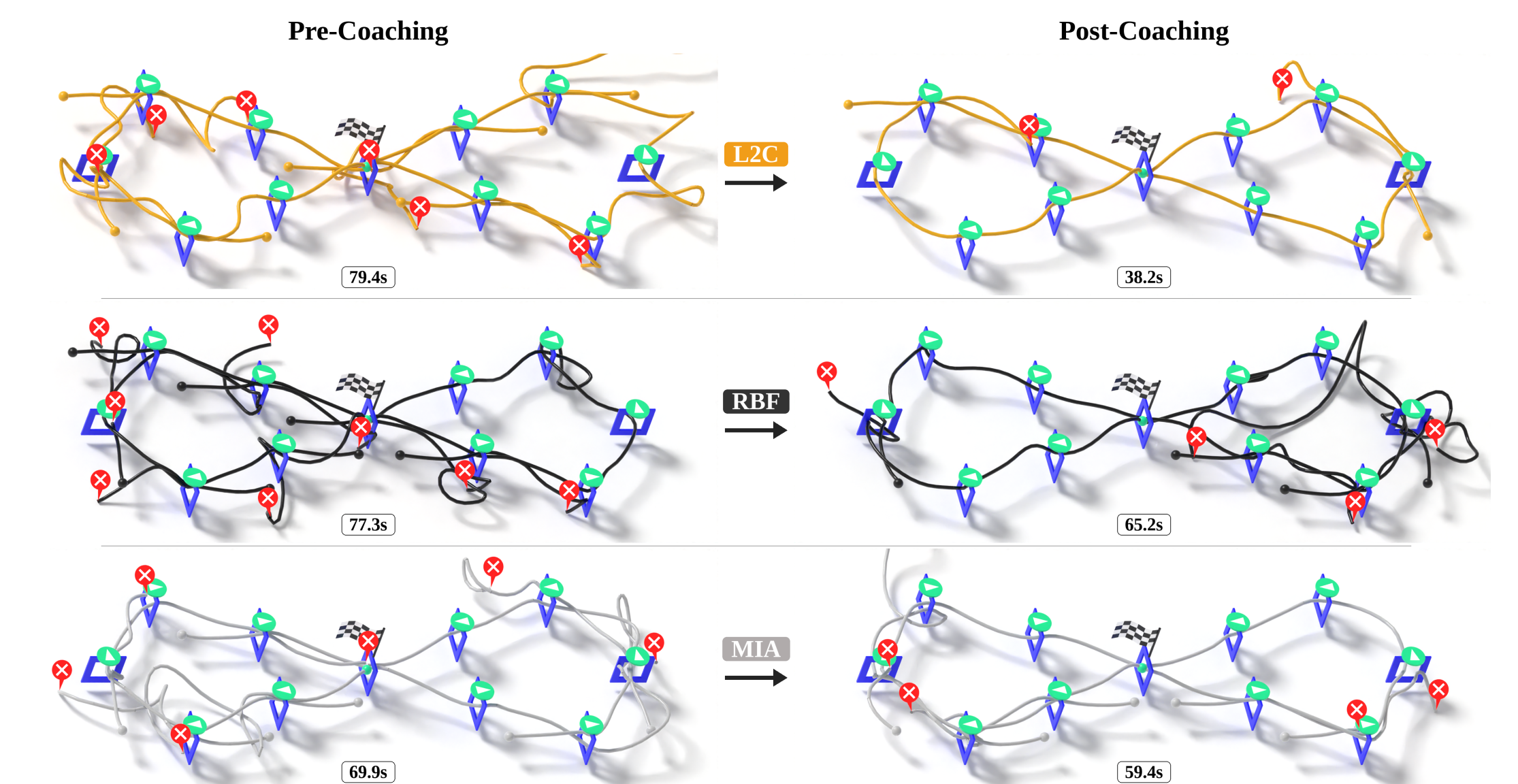
fewer failures per lap

$p < 0.001, d_z = -2.13$



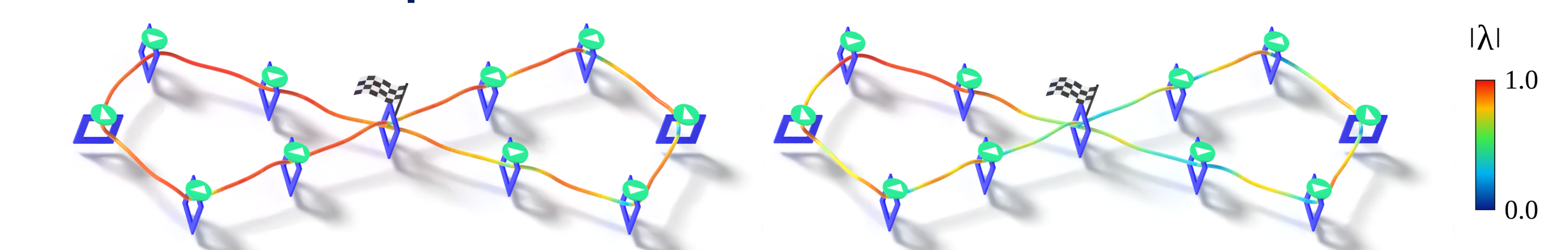
For L2C, a paired-samples test yielded a mean lap-time reduction of 27.9% with $p = 0.005$ and Cohen's $d_z = -1.08$. The mean failure count decreased by 3.52 failures per lap with $p < 0.001$ and $d_z = -2.13$.

Pre- and Post-Coaching Trajectories



Example pre- and post-coaching evaluation trajectories for the three coaching methods. Rows correspond to coaching methods; columns compare pre- and post-coaching evaluation laps.

Assistance Adaptation



L2C assistance coefficient $|\lambda|$ visualized on example racing trajectories. Left: estimated skill $\hat{\theta} = 0$; right: estimated skill $\hat{\theta} = 0.5$.

Limitations and Outlook

Although our probabilistic skill model appeared sufficient for drone racing, the simulated learners may not capture the full variability of real human learning in more complex motor-skill training settings.

